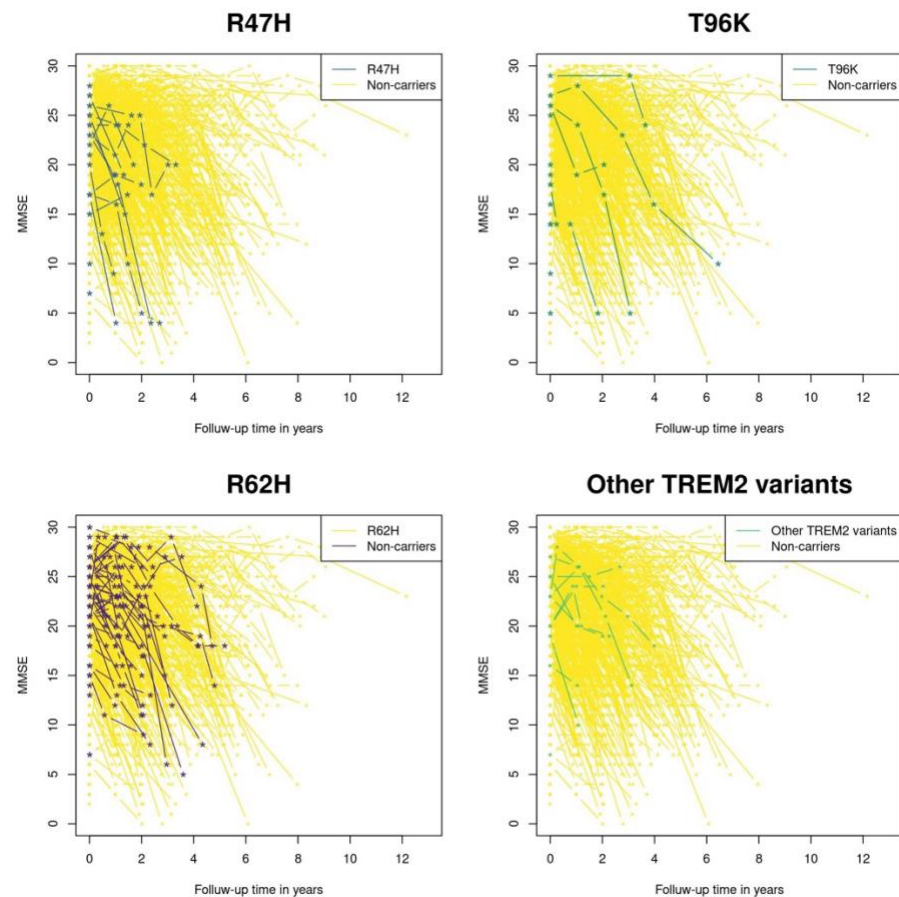


EXTENDED DATA

Extended Data Fig. 1 *TREM2* effect on cognitive decline stratified by *TREM2* mutation carriers compared to non-carriers.



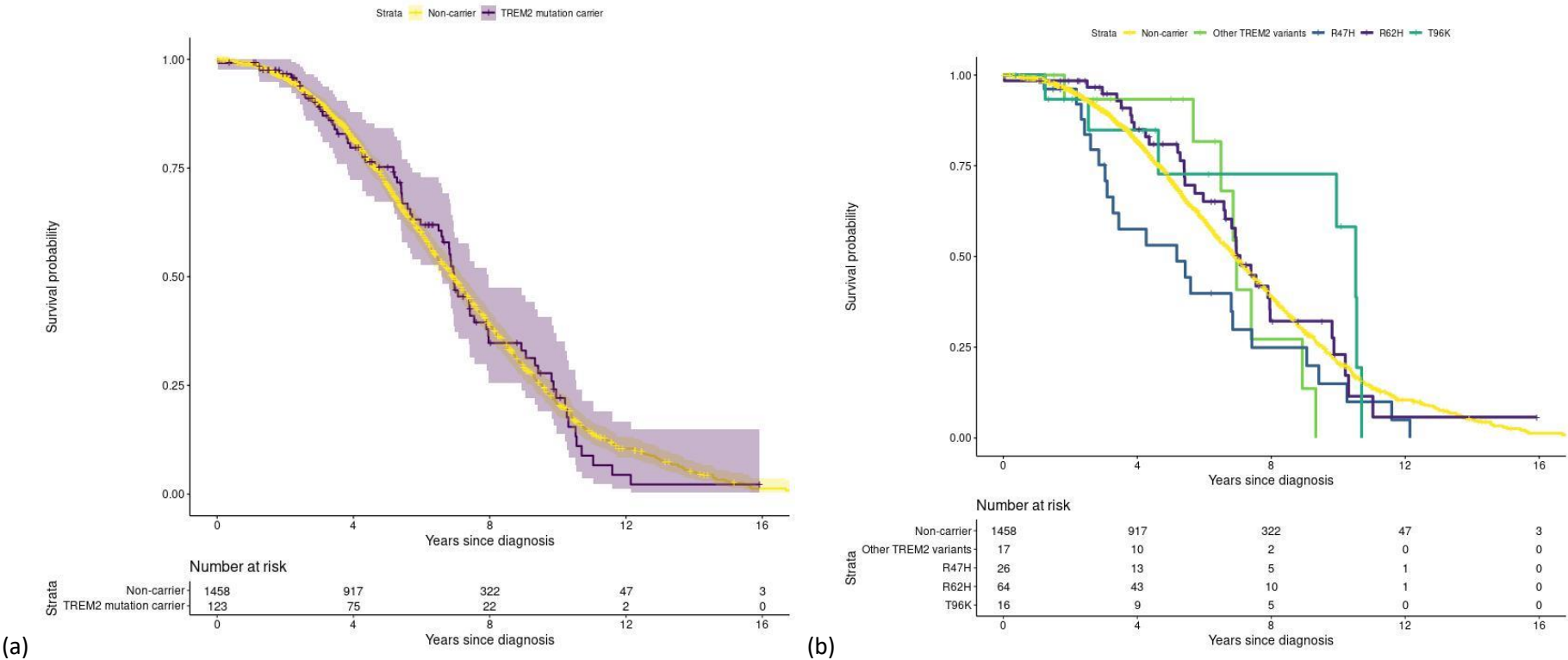
Shown here are MMSE values per individual over time, unadjusted.

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

Abbrev.: MMSE = Mini-Mental State Examination; R47H = p.Arg47His; R62H = p.Arg62His; T96K = p.Thr96Lys; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

SUPPLEMENTARY MATERIAL

Supplementary Fig. 1: *TREM2* effect on time between diagnosis of symptomatic AD and death compared to non-carriers.



Shown here are unadjusted Kaplan Meijer curves of the time between diagnosis and death.

(a) *TREM2* mutation carriers vs. non-carriers. (b) *TREM2*-mutation carriers stratified by mutation vs. non-carriers.

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

Abbreviations: R47H = p.Arg47His; R62H = p.Arg62His; T96K = p.Thr96Lys; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

Supplementary Table 1: Cohort characteristics per endophenotype.

	Main analysis								Exploratory analysis	
	Total	Neuropsych. Domains	MRI clinical ratings	CSF	CSF-NfL	EEG	Time Diagnosis to Death	MMSE over Time	MRI structural	Tau- PET
Total	1582 (100)	1519 (96)	1210 (76)	1522 (96)	164 (10)	1304 (82)	1582 (100)	1564 (99)	1069 (68)	67 (4)
Female	824 (52)	789 (52)	637 (53)	791 (52)	94 (57)	679 (52)	824 (52)	814 (52)	558 (52)	33 (49)
Age at diagnosis, <i>mean ±SD</i>	64.4 ±7	64.4 ±7	64.8 ±7	64.4 ±7	63.4 ±7	64.4 ±7.1	64.4 ±7	64.4 ±7	64.6 ±7	63 ±7
Education, <i>median (IQR)</i>	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	1 (0-2)	2 (1-2)
Positive family history ^a	691 (44)	670 (44)	524 (43)	660 (43)	78 (48)	565 (43)	691 (44)	688 (44)	466 (44)	35 (52)
ApoE-ε4 carrier	1076 (68)	1044 (69)	842 (70)	1038 (68)	115 (70)	899 (69)	1076 (68)	1064 (68)	743 (70)	46 (69)
Died	949 (60)	894 (59)	747 (62)	924 (61)	102 (62)	811 (62)	949 (60)	934 (60)	657 (61)	35 (52)
Age at death	71 ±8	71.1 ±8	71.5 ±8	71.1 ±7.7	69.2 ±8	71.2 ±8	71 ±8	71 ±8	71.2 ±8	69.4 ±7
<i>TREM2</i> mutation carriers	123 (8)	118 (8)	99 (8)	117 (8)	10 (6)	105 (8)	123 (8)	120 (8)	95 (9)	4 (6)
R62H	66 (54)	64 (54)	54 (55)	61 (52)	4 (40)	57 (54)	66 (54)	66 (55)	53 (56)	1 (25)
R47H	26 (21)	25 (21)	18 (18)	26 (22)	3 (30)	23 (22)	26 (21)	24 (20)	18 (19)	1 (25)
T96K	16 (13)	14 (12)	15 (15)	16 (14)	2 (20)	12 (11)	16 (13)	15 (12)	11 (12)	0 (0)
Other variants*	17 (14)	17 (14)	13 (13)	16 (14)	1 (10)	15 (14)	17 (14)	17 (14)	14 (15)	2 (50)

Table shows *n* (%) unless otherwise specified.

*Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

a) Positive family history = affected first degree relative.

Abbreviations: AD = Alzheimer's Disease; ApoE = Apolipoprotein E; CSF = cerebrospinal fluid; Diag. = Diagnosis; EEG = Electroencephalography; IQR = Interquartile Range; MMSE = Mini-Mental State Examination; MRI = Magnetic Resonance Imaging; n = number; NfL = Neurofilament Light; Neuropsych. = Neuropsychological; PET = Positron Emission Tomography; R47H = p.Arg47His; R62H = p.Arg62His; T96K = p.Thr96Lys; SD = Standard Deviation; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

Supplementary Table 2: Effect of *TREM2* variants on clinical measures in AD comparing *TREM2* mutation carriers vs. non-carriers, scaled.

	Main analysis		Exploratory analysis							
	All <i>TREM2</i> variants combined		R62H		R47H		T96K		Other <i>TREM2</i> variants	
	stdβ ± SE	p-value [★]	stdβ ± SE	p-value	stdβ ± SE	p-value	stdβ ± SE	p-value	stdβ ± SE	p-value
MMSE at baseline	-0.00 ±0.08	0.962	0.07 ±0.11	0.544	-0.11 ±0.18	0.54	-0.19 ±0.20	0.339	0.08 ±0.22	0.724
<u>Neuropsychological domains</u>										
Memory	-0.14 ±0.07	0.247	-0.10 ±0.10	0.327	-0.27 ±0.16	0.097	-0.42 ±0.18	0.018*	0.13 ±0.19	0.487
Executive functioning	0.07 ±0.16	0.816	0.21 ±0.21	0.316	-0.69 ±0.35	0.053	-0.36 ±0.37	0.331	0.63 ±0.39	0.110
Attention and speed	0.12 ±0.10	0.667	0.28 ±0.14	0.042*	-0.40 ±0.22	0.072	0.18 ±0.25	0.463	0.07 ±0.25	0.774
Language	-0.03 ±0.08	0.816	0.18 ±0.10	0.075	-0.38 ±0.17	0.027*	-0.37 ±0.18	0.034*	-0.00 ±0.19	0.996
Visuospatial functioning	0.21 ±0.08	0.094 [★]	0.16 ±0.10	0.12	0.24 ±0.17	0.174	0.24 ±0.18	0.197	0.29 ±0.19	0.123
Global cognition	-0.04 ±0.07	0.816	0.02 ±0.08	0.824	-0.56 ±0.16	0.0004***	-0.15 ±0.15	0.329	0.28 ±0.16	0.076
<u>Magnetic resonance imaging</u>										
Medial Temporal Atrophy	-0.06 ±0.10	0.816	-0.26 ±0.13	0.052	0.07 ±0.23	0.771	0.31 ±0.21	0.144	0.09 ±0.27	0.731
Parietal Cortical Atrophy	-0.10 ±0.11	0.667	0.13 ±0.14	0.343	-0.32 ±0.24	0.181	-0.56 ±0.23	0.013*	-0.43 ±0.28	0.128
Fazekas score	-0.14 ±0.10	0.479	-0.27 ±0.13	0.035*	0.03 ±0.22	0.903	0.03 ±0.20	0.875	0.10 ±0.26	0.685
<u>Cerebrospinal fluid</u>										
Aβ42	-0.09 ±0.10	0.667	-0.04 ±0.13	0.760	0.21 ±0.20	0.297	-0.64 ±0.21	0.003**	-0.18 ±0.25	0.475
pTau-181	0.03 ±0.10	0.847	0.00 ±0.13	0.994	0.60 ±0.20	0.003**	-0.50 ±0.21	0.019*	-0.04 ±0.25	0.877
t-Tau	0.05 ±0.10	0.816	0.08 ±0.13	0.557	0.47 ±0.20	0.018*	-0.45 ±0.21	0.035*	-0.05 ±0.25	0.833
Neurofilament Light	0.08 ±0.32	0.797	-0.02 ±0.51	0.963	0.35 ±0.59	0.556	0.51 ±0.71	0.475	-1.05 ±1.00	0.297
	β ± SE	p-value [★]	β ± SE	p-value	β ± SE	p-value	β ± SE	p-value	β ± SE	p-value
<u>EEG abnormality</u> ^{log}	-0.44 ±0.21	0.227	-0.38 ±0.29	0.195	-0.21 ±0.43	0.633	-0.62 ±0.55	0.369	-0.96 ±0.60	0.133
<u>Time diagnosis-death</u> ^{cox}	0.12 ±0.13	0.667	0.05 ±0.18	0.759	0.47 ±0.22	0.034*	-0.42 ±0.41	0.311	0.13 ±0.36	0.706
<u>Cognitive decline</u> ^{lmm}	-0.63 ±0.25	0.094 [★]	-0.54 ±0.31	0.085	-1.40 ±0.61	0.021*	-1.75 ±0.74	0.018*	0.58 ±0.62	0.350

Shown here are *standardized* betas of linear regression models adjusted for age and sex; MMSE at baseline and neuropsychological domains are also adjusted for education level and disease stage. Also shown here are betas of logistic regression models (log), cox proportional hazards models (cox), and linear mixed models (lmm) adjusted for age, sex, education level, and disease stage; cox regression models are also adjusted for MMSE at baseline.

◆ FDR-corrected $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

Abbreviations: A β 42 = Beta-Amyloid 42; CI = confidence interval; EEG = Electroencephalography; EF = Executive Functioning; FDR = False Discovery Rate; MMSE = Mini-Mental State Examination; pTau-181 = Phosphorylated Tau-181; R47H = p.Arg47His; R62H = p.Arg62His; SE = Standard Error; std β = standardized beta; T96K = p.Thr96Lys; *TREM2* = Triggering Receptor Expressed On Myeloid Cells 2.

Supplementary Table 3: Effect of *TREM2* variants on time between diagnosis and death.

	Hazard Ratio [95%-Confidence Interval]	p-value	p-value FDR-corrected
Model 1			
All <i>TREM2</i> variants	1.12 [0.9-1.4]	0.353	0.667
Model 2			
R62H	1.06 [0.8-1.5]	0.759	NA
R47H	1.60 [1.0-2.5]	0.034*	NA
T96K	0.66 [0.3-1.5]	0.311	NA
Other <i>TREM2</i> variants	1.14 [0.6-2.3]	0.706	NA

Shown here are hazard ratios with 95% confidence interval, adjusted for age at diagnosis, sex, education level, disease stage (MCI/dementia), and MMSE at baseline.

*p<0.05

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

Abbreviations: NA = Not Available; FDR = False Discovery Rate; R47H = p.Arg47His; R62H = p.Arg62His; T96K = p.Thr96Lys; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

Supplementary Table 4: Summary statistics of quantitative image analysis of significant *TREM2*-associated cortical thickness regions and subcortical volumes based on Desikan Kiliany atlas.

	Lobe	Brain region of interest	std β $\pm$ SE	p-value
All <i>TREM2</i> variants	Subcortical	Amygdala	0.19 $\pm$ 0.10	0.047
R62H	Temporal	Temporal pole	0.29 $\pm$ 0.14	0.037
R47H	Occipital	Cuneus	0.49 $\pm$ 0.24	0.037
		Lingual	0.47 $\pm$ 0.24	0.049
	Subcortical	Amygdala	0.69 $\pm$ 0.21	0.001
		Hippocampus	0.49 $\pm$ 0.20	0.016
T96K	Cingulate	Rostral anterior cingulate	-0.67 $\pm$ 0.25	0.007
	Frontal	Medial orbitofrontal	-0.50 $\pm$ 0.25	0.047
	Subcortical	Hippocampus	-0.64 $\pm$ 0.22	0.003
	NA	Insula	-0.55 $\pm$ 0.25	0.029
Other <i>TREM2</i> variants	Cingulate	Isthmus cingulate	-0.61 $\pm$ 0.28	0.027
	Frontal	Pars orbitalis	-0.59 $\pm$ 0.27	0.033
		Frontal pole	-0.56 $\pm$ 0.28	0.042

Shown here are standardized betas with standard error, adjusted for age, sex and estimated intracranial volume. The betas represent the effect of *TREM2* variants on cortical thickness (mm) and subcortical volumes (mm³) compared to non-carriers with symptomatic AD.

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R47G, R31F, and A105V.

Abbreviations: CI = confidence interval; R47H = p.Arg47His; T96K = p.Thr96Lys; SE = Standard Error; std β = standardized beta; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

Supplementary Table 5: Sensitivity analyses of *TREM2* effect on clinical measures in symptomatic Alzheimer's disease, scaled.

Shown here are (standardized) betas of the sensitivity analyses. (a) Cohort descriptives per clinical measure. (b) Shown here are standardized betas of linear regression models are adjusted for age, sex, and three principal components. Neuropsychological domains are also adjusted for education level and disease stage (MCI/dementia). Also shown here are betas of logistic regression models (log), cox proportional hazards models (cox), and linear mixed models (lmm) adjusted for age, sex, education level, disease stage and three principal components; cox regression models are also adjusted for MMSE at baseline. (c) Cox regression models are adjusted for age at diagnosis, sex, education level, disease stage, MMSE at baseline, and three principal components. (d) Significant findings of linear regression models in quantitative image analysis adjusted for age, sex, estimated intracranial volume, and three principal components ($p < 0.05$).

◆ FDR-corrected $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Other *TREM2* variants include: D87N, G58A, Q33X, C51Y, R31F.

Abbreviations: A β 42 = Beta-Amyloid 42; AD = Alzheimer's Disease; ApoE = Apolipoprotein E; CI = confidence interval; CSF = cerebrospinal fluid; EF = Executive Functioning; IQR = Interquartile Range; MMSE = Mini-Mental State Examination; MRI = Magnetic Resonance Imaging; MTA = Medial Temporal Lobe; n = number; NA = Not Applicable; NfL = Neurofilament Light; Neuropsych. = Neuropsychological; PCA = Posterior Cortical Atrophy; pTau-181 = Phosphorylated Tau-181; R47H = p.Arg47His; R62H = p.Arg62His; SD = Standard Deviation; SE = Standard Error; std β = standardized beta; T96K = p.Thr96Lys; *TREM2* = Triggering Receptor Expressed on Myeloid Cells 2.

a)

	Main analysis								Exploratory analysis	
	Total	Neuropsych. Domains	MRI clinical ratings	CSF	CSF-NfL	EEG	Time Diagnosis to Death	MMSE over Time	MRI structural	Tau-PET
Total	1444 (100)	1391 (96)	1102 (76)	1390 (96)	150 (10)	1189 (82)	1444 (100)	1431 (99)	968 (67)	63 (4)
Female	747 (52)	717 (52)	572 (52)	719 (52)	85 (57)	616 (52)	747 (52)	739 (52)	496 (51)	31 (49)
<i>TREM2</i> mutation carriers	103 (7)	101 (7)	84 (8)	97 (7)	8 (5)	89 (7)	103 (7)	101 (7)	83 (9)	4 (6)
R62H	60 (58)	59 (58)	51 (61)	55 (57)	4 (50)	52 (58)	60 (58)	60 (59)	50 (60)	1 (25)
R47H	26 (25)	25 (25)	18 (21)	26 (27)	3 (38)	23 (26)	26 (25)	24 (24)	18 (22)	1 (25)
T96K	3 (3)	3 (3)	3 (4)	3 (3)	0 (0)	2 (2)	3 (3)	3 (3)	2 (2)	0 (0)
Other variants*	16 (16)	16 (16)	13 (15)	15 (15)	1 (12)	14 (16)	16 (16)	16 (16)	14 (17)	2 (50)

Table shows n (%).

b)

	Main analysis		Exploratory analysis							
	All <i>TREM2</i> variants combined		R62H		R47H		T96K		Other <i>TREM2</i> variants	
	stdβ ± SE	p-value [★]	stdβ ± SE	p-value	stdβ ± SE	p-value	stdβ ± SE	p-value	stdβ ± SE	p-value
MMSE at baseline	0.09 ±0.10	0.555	0.08 ±0.13	0.506	-0.14 ±0.20	0.497	0.45 ±0.56	0.419	0.31 ±0.24	0.196
<u>Neuropsychological domains</u>										
Memory	-0.08 ±0.08	0.555	-0.08 ±0.10	0.443	-0.29 ±0.17	0.079	-0.43 ±0.45	0.340	0.22 ±0.20	0.270
Executive functioning	0.16 ±0.17	0.555	0.26 ±0.22	0.233	-0.69 ±0.35	0.052	-0.46 ±0.91	0.612	0.81 ±0.40	0.046*
Attention and speed	0.08 ±0.11	0.700	0.25 ±0.14	0.082	-0.40 ±0.22	0.072	0.58 ±0.60	0.337	0.06 ±0.26	0.832
Language	-0.01 ±0.08	0.943	0.15 ±0.10	0.153	-0.39 ±0.17	0.022*	-0.36 ±0.45	0.430	0.02 ±0.20	0.904
Visuospatial functioning	0.22 ±0.08	0.136	0.19 ±0.11	0.072	0.23 ±0.17	0.188	0.20 ±0.41	0.629	0.30 ±0.19	0.122
Global cognition	-0.03 ±0.07	0.872	0.03 ±0.09	0.765	-0.56 ±0.16	0.0004***	-0.18 ±0.34	0.589	0.27 ±0.16	0.082
<u>Magnetic resonance imaging</u>										
Medial Temporal Atrophy	-0.12 ±0.11	0.555	-0.23 ±0.14	0.099	0.06 ±0.23	0.783	0.00 ±0.56	0.998	0.08 ±0.27	0.770
Parietal Cortical Atrophy	-0.02 ±0.11	0.943	0.19 ±0.14	0.189	-0.34 ±0.24	0.150	-0.45 ±0.71	0.525	-0.44 ±0.28	0.116
Fazekas score	-0.18 ±0.11	0.353	-0.30 ±0.13	0.025*	0.02 ±0.22	0.914	-0.58 ±0.54	0.281	0.08 ±0.26	0.755
<u>Cerebrospinal fluid</u>										
Aβ42	0.02 ±0.10	0.943	-0.00 ±0.14	0.994	0.22 ±0.20	0.273	-0.52 ±0.57	0.365	-0.17 ±0.26	0.498
pTau-181	0.11 ±0.10	0.555	-0.00 ±0.14	0.989	0.60 ±0.20	0.003**	-0.63 ±0.58	0.276	-0.05 ±0.26	0.839
t-Tau	0.11 ±0.10	0.555	0.08 ±0.14	0.548	0.46 ±0.20	0.020*	-0.74 ±0.58	0.199	-0.09 ±0.26	0.735
Neurofilament Light	-0.06 ±0.37	0.865	-0.04 ±0.52	0.931	0.28 ±0.60	0.638	NA	NA	NA	NA
	β ± SE	p-value [★]	β ± SE	p-value	β ± SE	p-value	β ± SE	p-value	β ± SE	p-value
<u>EEG abnormality</u> ^{log}	-0.49 ±0.23	0.249	-0.37 ±0.30	0.205	-0.21 ±0.43	0.631	-12.9 ±NA	0.969	-1.2 ±0.67	0.065
<u>Time diagnosis-death</u> ^{cox}	0.14 ±0.13	0.555	-0.01 ±0.18	0.951	0.46 ±0.22	0.038*	0.08 ±1.0	0.934	0.08 ±0.38	0.844
<u>Cognitive decline</u> ^{lmm}	-0.52 ±0.26	0.249	-0.61 ±0.32	0.054	-1.38 ±0.61	0.022*	0.26 ±1.2	0.833	0.61 ±0.63	0.332

c)

	Hazard Ratio [95% Confidence Interval]	p-value	p-value FDR-corrected
All <i>TREM2</i> variants	1.15 [0.9-1.5]	0.289	0.555
R62H	0.99 [0.7-1.4]	0.951	NA
R47H	1.59 [1.0-2.5]	0.038*	NA
T96K	1.09 [0.1-7.8]	0.934	NA
Other <i>TREM2</i> variants	1.08 [0.5-2.3]	0.844	NA

d)

	Lobe	Brain region of interest	std β $\pm$ SE	p-value
All <i>TREM2</i> variants	Subcortical	Amygdala	0.21 $\pm$ 0.10	0.039
R47H	Occipital	Cuneus	0.48 $\pm$ 0.24	0.041
	Subcortical	Amygdala	0.70 $\pm$ 0.21	0.001
		Hippocampus	0.47 $\pm$ 0.20	0.023
T96K	Temporal	Parahippocampal	1.55 $\pm$ 0.69	0.025
Other <i>TREM2</i> variants	Cingulate	Isthmus cingulate	-0.61 $\pm$ 0.28	0.028
	Frontal	Pars orbitalis	-0.55 $\pm$ 0.28	0.046